# Hongwen Pu

Year 2, MEng Mathematical Computation (Computer Science and Mathematics)
University College London
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## Research Interests Interested in

- Mathematical foundations of machine learning
- Safety and alignment of agentic AI systems
- AI model verification and safety
- Algorithmic ethics and policy

## Publications

Pu, H., Yi, K. (2024).
*A Comparative Analysis of EfficientNet and MobileNet Models' Performance on Limited Datasets: An Example of American Sign Language Alphabet Detection.*
**Highlights in Science, Engineering and Technology**, 94, 558–564.
DOI: 10.54097/yh5d3s04.

- Compared two popular lightweight CNN architectures - EfficientNet and MobileNet under limited-data constraints.
- Evaluated performance across 5,10, and 20 training epochs on merged ASL datasets with sources taken from Kaggle
- Observed an early-epoch performance advantage of MobileNetV2 and systematic overfitting behaviour beyond 10 epochs.

*Accepted by the 2024 2nd International Conference on Computer, Machine Learning and Artificial Intelligence (CMLAI 2024); 8 citations (Google Scholar); 496 publisher-page downloads as of Nov 2025.*

## Research Experience

**Biomedical NLP Systematic Review** 2026–Present

- Contributing to a systematic review in Biomedical NLP as an independent second reviewer.
- Participating in the review process as the project progresses towards completion and manuscript preparation.

**American Sign Language Spelling Recognition System and Educational Game (2024)**

- Developed a MobileNet-based ASL alphabet recognition system using a low-code platform, Edge Impulse
- Deployed the trained model to a web-based interactive learning game through Edge Impulse SDK with Vue.js
- Achieved the highest overall grade among all project groups

Role: Model training, frontend development, deployment, final viva presentation.

**MVITA: Multi-Cultural Virtual Interview Agent (Hackathon Project, 2025)**

- Developed a web-based virtual interview system for cross-cultural job preparation through GPT4.5 API and GPT4.5-based Anam SDK

- Implemented the frontend using Vue.js and the Anam SDK; backend using Python (Flask).
- Integrated GPT API for interview question generation in cultural contexts.

Role: Frontend development, SDK integration, prompt engineering.

## Education

**University College London** London, UK
MEng Mathematical Computation (Computer Science and Mathematics) 2025–Present
*Currently in Year 2*
**Year 1 Average:** First-Class level

**Relevant Coursework:** Analysis, Algebra, Security, Theory of Computation, Probability, Algorithms, Mathematical Methods, Directed Reading

**Dulwich College (Singapore)** Singapore, Singapore
IB Diploma (Bilingual) 2023–2025
HL: Mathematics AA (7), Computer Science (7), Physics (6)

## Skills

Programming: Python, JavaScript/TypeScript, Vue.js, Node.js, Java, C, Haskell
Tools: LaTeX, Git
Languages: Chinese (Native), English (C2), Japanese (B1–B2)

## Teaching Experience

**UCL Chinese Students and Scholar Association (CSSA)**
Head of Department, Department of Chinese Education 2026–Present

- Overseeing and coordinating the classes under the Department of Chinese Education
- Coordinating and organising activities across departments that lead to cultural exchange and communication in the school
- Worked with multiple stakeholders such as the Student Union UCL, sponsors and other departments within CSSA, and am familiar with their procedures.

Higher-Level Chinese Instructor, Department of Chinese Education 2025–2026

- Designed and delivered advanced Chinese Language and Literature courses to second-language students
- Led close reading and literary analysis of "Taipei People", allowing a greater understanding of Chinese culture
- Integrated linguistic, cultural and historical perspectives into the course

**Dulwich College (Singapore)**
HiMCM Coordinator & Mathematical Modelling Teaching Lead 2023–2025

- Coordinated students' participation at DCSG in the HiMCM competition - a competition that tests students' ability to construct mathematical models and work collaboratively
- Taught modelling methods, including AHP and TOPSIS, with an emphasis on conceptual understanding